The best way to think about renormalization wasn't really appreciated until the work of Nobel laureate Kenneth Wilson in the early 1970s. Wilson realized that all of the infinities in Feynman-diagram calculations came from virtual particles with very large energies, corresponding to processes at extremely short distances. But high energies and short distances are precisely where we should have the least confidence that we know what's going on. Processes with very high energies could involve completely new fields, ones that have such high masses that we haven't yet produced them in experiments. For that matter, spacetime itself might break down at short distances, perhaps at the Planck length.

So, Wilson reasoned, what if we're just a little bit more honest, and admit that we don't know what's going on at arbitrarily high energies? Instead of taking loops in Feynman diagrams and allowing the energies of the virtual particles to go up to infinity, let's include an explicit cutoff in the theory: an energy above which we don't pretend to know what's happening. The cutoff is in some sense arbitrary, but it makes sense to put it at the dividing line between energies about which we have good experimental knowledge, and above which we haven't been able to peek. There can even be a physically good reason to choose a certain cutoff, if we expect new particles or other phenomena to kick in at that scale, but don't know exactly what they will be.

